Applied Data Mining

— Week 6—

Spring 2026

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Outline

- Decision Tree Induction

Decision Tree Induction

- It is the learning of decision trees from class-labeled training tuples.

- A decision tree is a flowchart-like tree structure, where each internal node (non-leaf node) denotes a test on an attribute, and each branch represents an outcome of the test.

- Each leaf node (or terminal node) holds a class label. The topmost node in a tree is the root node.

Decision Tree Induction

"How are decision trees used for classification?"

•Given a tuple, ***X***, for which the associated class label is unknown.

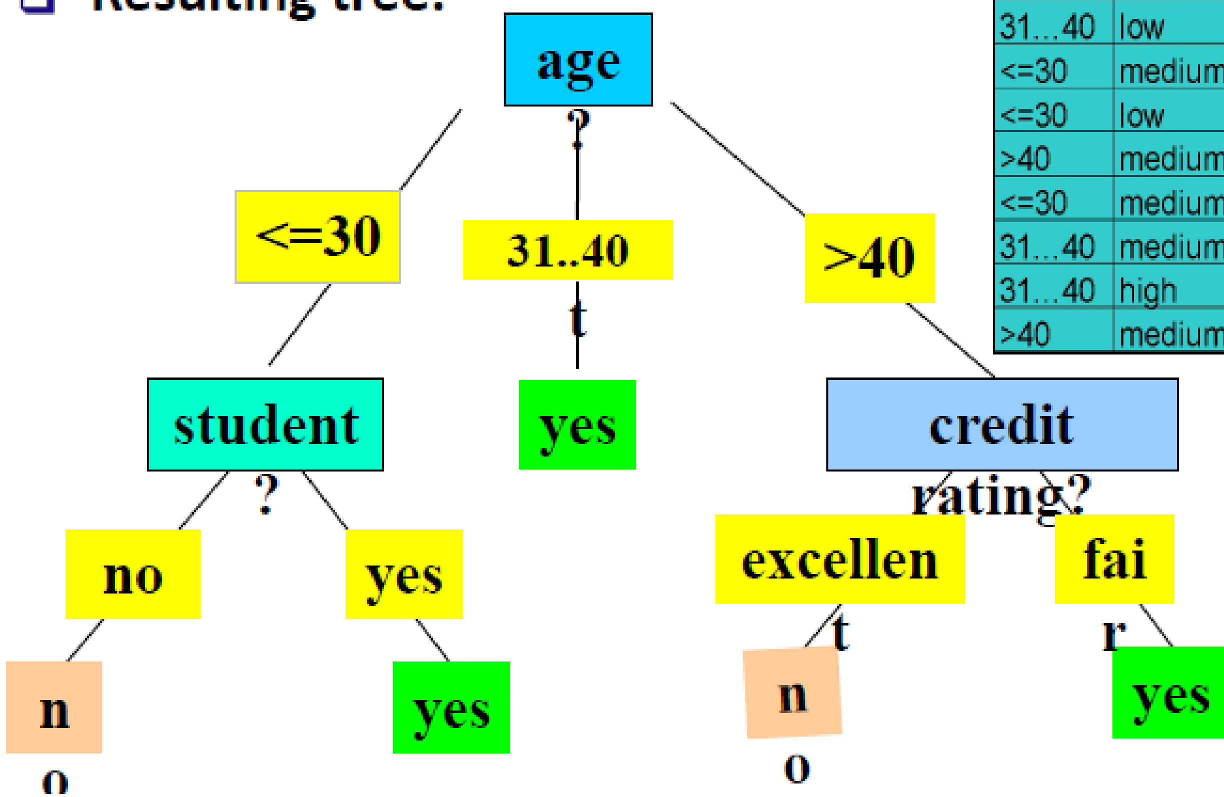
•The attribute values of the tuple are tested against the decision tree.

•A path is traced from the root to a leaf node, which holds the class prediction for that tuple.

•Decision trees can easily be converted to classification rules.

Decision Tree Induction: An Example

- Training data set: Buys_computer
- The data set follows an example of Quinlan's ID3 (buys_computer)
- Resulting tree:



age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

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Steps for Decision Tree Induction

Basic algorithm steps (ID3)

– Tree is constructed in a top-down recursive divide-and-conquer manner

– At start, all the training examples are at the root

– Attributes are categorical (if continuous-valued, they are discretized in advance)

– Examples are partitioned recursively based on selected Attributes

– Test attributes are selected on the basis of a heuristic or statistical measure (e.g., information gain)

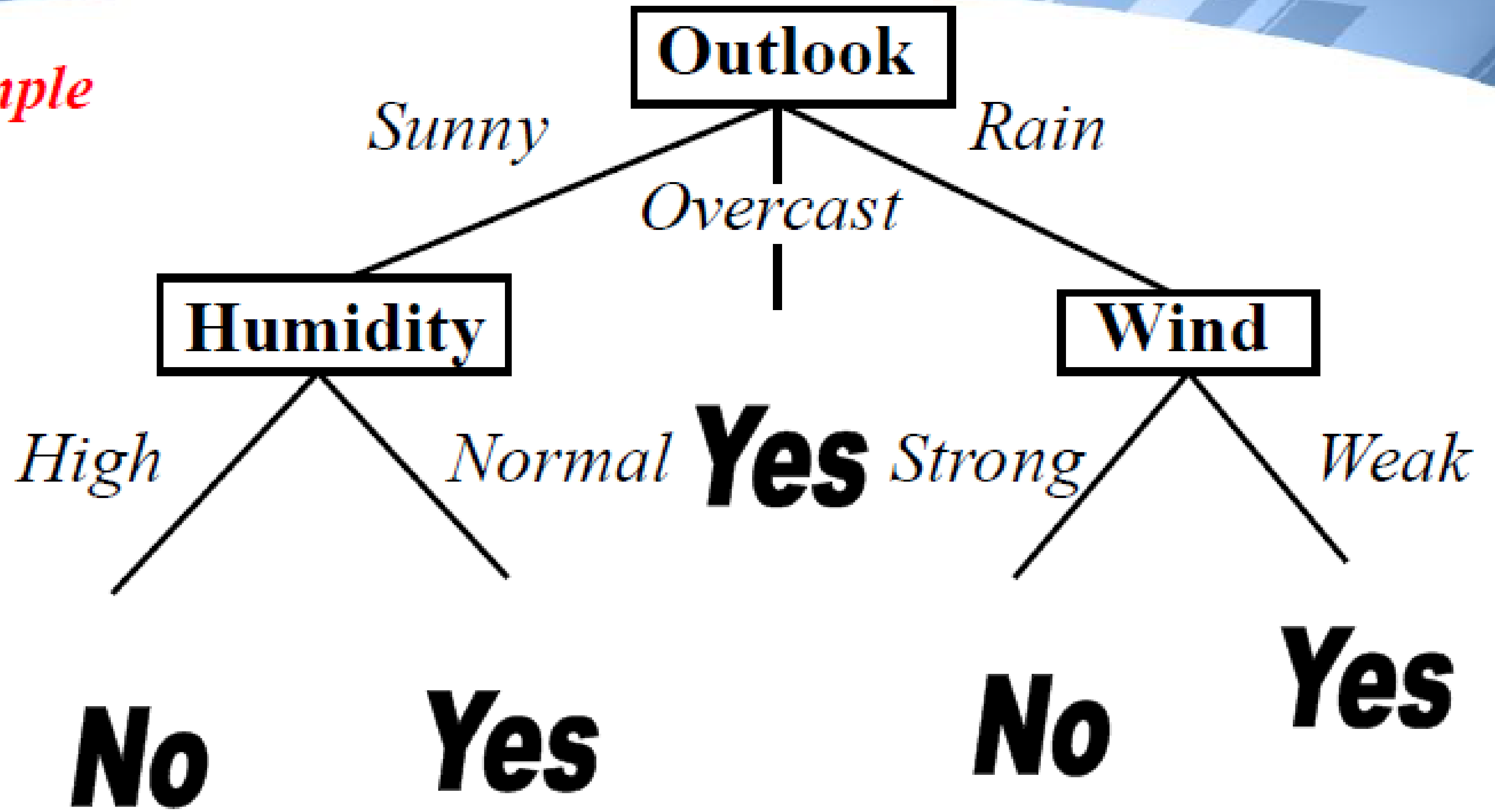
Decision Tree

Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Decision Tree

Example



A Decision Tree for the concept *PlayTennis*

An unknown observation is classified by testing its attributes and reaching a leaf node

Decision Tree

Decision Tree Representation

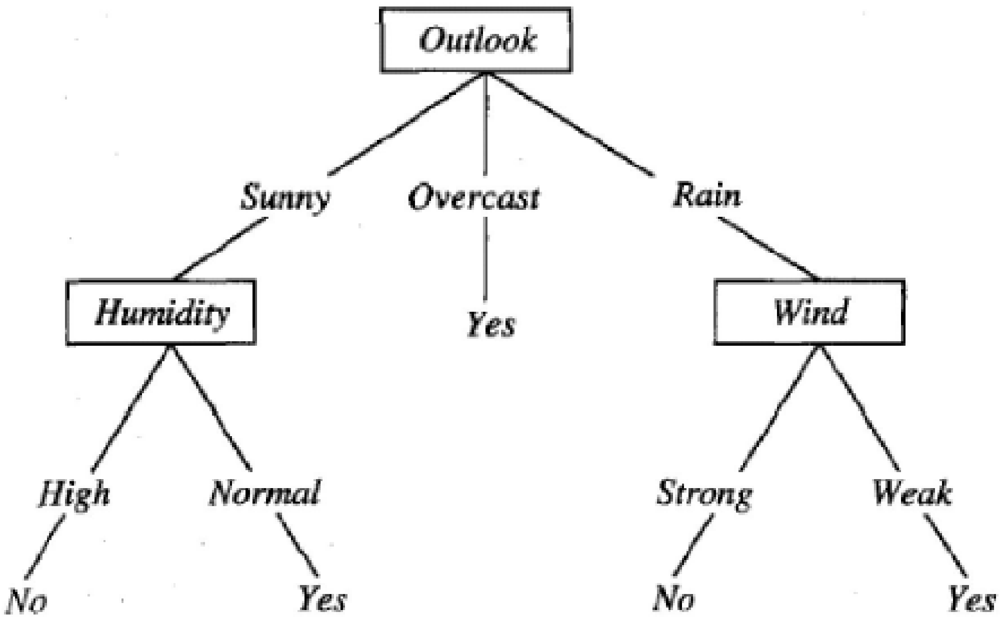
Decision trees represent a disjunction of conjunctions of constraints on the attribute values of instances

Each path from the tree *root* to a *leaf* corresponds to a conjunction of attribute tests (one rule for classification)

The tree itself corresponds to a disjunction of these conjunctions (set of rules for classification)

Decision Tree

Decision Tree Representation



$$
\begin{aligned}
& (Outlook = Sunny \wedge Humidity = Normal) \\
\vee\ & (Outlook = Overcast) \\
\vee\ & (Outlook = Rain \wedge Wind = Weak)
\end{aligned}
$$

Decision Tree

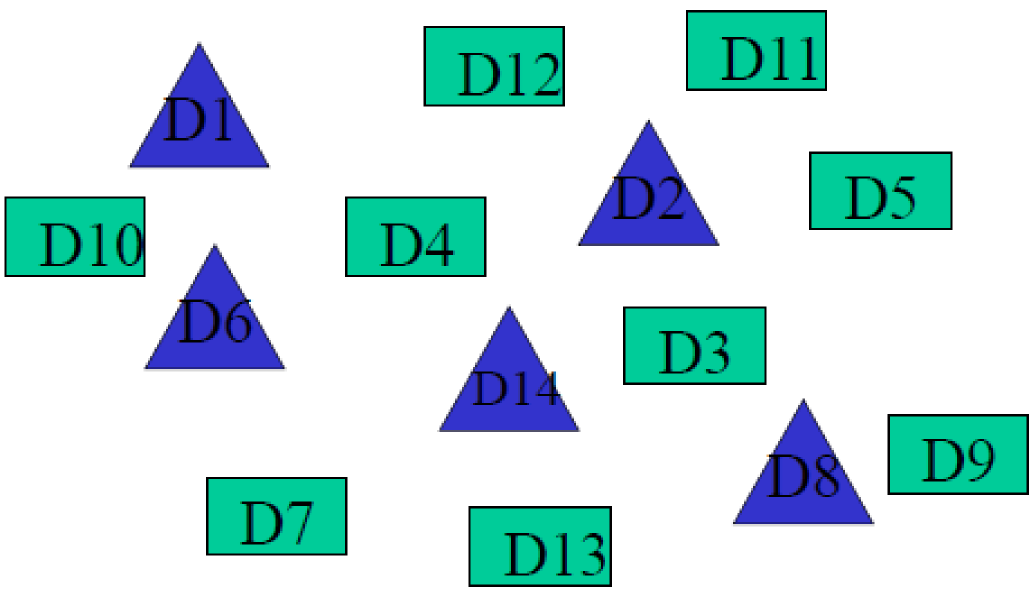
Basic Decision Tree Learning Algorithm

First of all, we *select* the best attribute to be tested at the root of the tree

To do this selection, each attribute is evaluated using a statistical test to determine how well it alone classifies the training examples

Decision Tree

Basic Decision Tree Learning Algorithm



We have

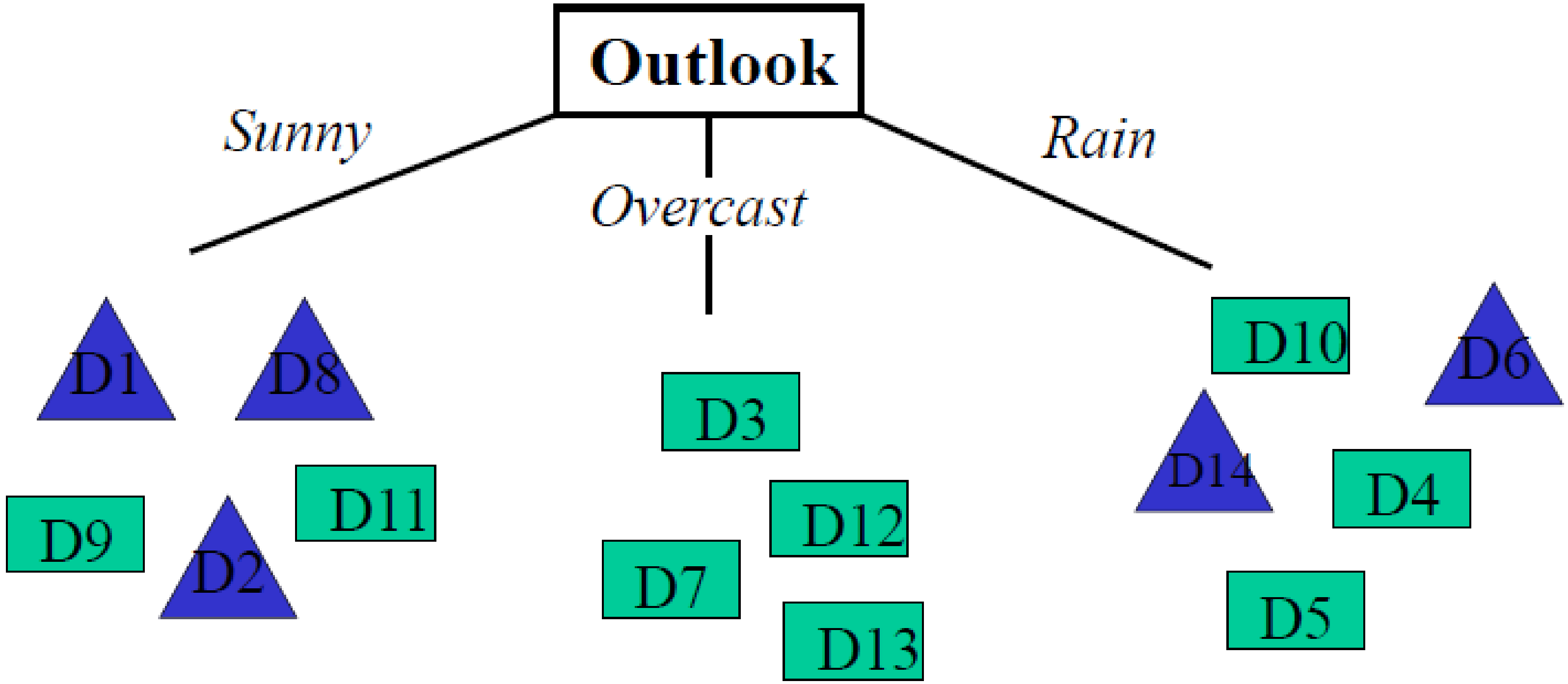
- 14 observations

- 4 attributes
- **Outlook**
- **Temperature**
- **Humidity**
- **Wind**

- 2 classes (Yes, No)

Decision Tree

Basic Decision Tree Learning Algorithm

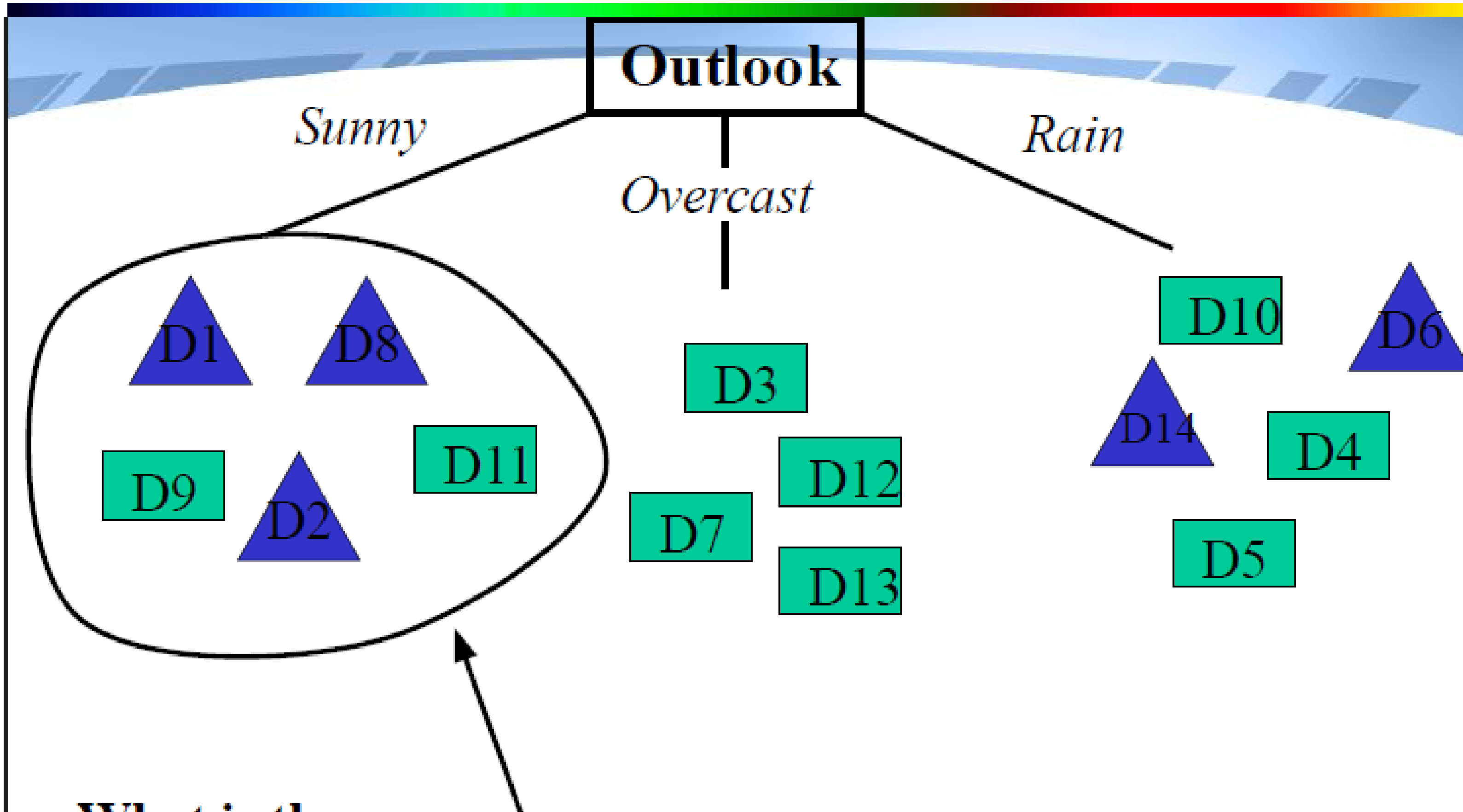


Decision Tree

Basic Decision Tree Learning Algorithm

The selection process is then repeated using the training examples associated with each descendant node to select the best attribute to test at that point in the tree

Decision Tree



What is the "best" attribute to test at this point? The possible choices are Temperature, Wind & Humidity

Decision Tree

Which Attribute is the Best Classifier?

The central choice in the ID3 algorithm is selecting which attribute to test at each node in the tree

We would like to select the attribute which is most useful for classifying examples

For this we need a good quantitative measure

For this purpose a statistical property, called *information gain* is used

Decision Tree

Which Attribute is the Best Classifier? Definition of Entropy

In order to define information gain precisely, we begin by defining entropy

Entropy is a measure commonly used in information theory

Entropy characterizes the impurity of an arbitrary collection of examples

OR

Entropy measures homogeneity of examples

Decision Tree

Which Attribute is the Best Classifier? Definition of Entropy

Suppose X can have *m* values, V_1, V_2, …, V_m, with probabilities: p_1, p_2, …, p_m

If the target attribute can take on m different values, then the entropy of *S* relative to this m-wise classification is:

$$Entropy(S) = H(S) = -\sum_{j=1}^{m} p_j \log_2 p_j$$

Where *pj* is the proportion of *S* belonging to class *j*. If a p = 1 and all other p's are 0, then entropy will be zero

Decision Tree

Which Attribute is the Best Classifier? Definition of Entropy

Suppose S is a collection of 14 examples of some boolean concept, including 9 positive and 5 negative examples

We adopt the notation [9+, 5-] to summarize such a sample of data).

Then the entropy of S relative to this boolean classification is

$$Entropy([9+, 5-]) = -(9/14)\log_2(9/14) - (5/14)\log_2(5/14)$$
$$= 0.940$$

Decision Tree

Which Attribute is the Best Classifier? Information Gain

Information Gain is the expected reduction in entropy caused by partitioning the examples according to an attribute's value

Info Gain (Y | A) = H(Y) – H(Y | A)

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

In general, we can write Gain (S, A), Where S is the collection of examples & A is an attribute.

Decision Tree

Which Attribute is the Best Classifier? Information Gain

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
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D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Let's investigate the attribute *Wind*

Decision Tree

Which Attribute is the Best Classifier? Information Gain

The collection of examples has 9 positive values and 5 negative ones

$$Entropy(S) = 0.940$$

Eight (6 positive and 2 negative ones) of these examples have the attribute value *Wind* = *Weak*

Six (3 positive and 3 negative ones) of these examples have the attribute value *Wind* = *Strong*

Decision Tree

Which Attribute is the Best Classifier? Information Gain

The information gain obtained by separating the examples according to the attribute *Wind* is calculated as:

$$
\begin{aligned}
Gain(S, Wind) &= Entropy(S) - \sum_{v \in \{Weak, Strong\}} \frac{|S_v|}{|S|} Entropy(S_v) \\
&= Entropy(S) - (8/14) Entropy(S_{Weak}) \\
&\quad - (6/14) Entropy(S_{Strong}) \\
&= 0.940 - (8/14) 0.811 - (6/14) 1.00 \\
&= 0.048
\end{aligned}
$$

Decision Tree

Which Attribute is the Best Classifier? Information Gain

$$\text{Entropy}\ (S_{Weak}) = -\,6/8\ \log_2 6/8 \quad -\,2/8\ \log_2 2/8$$

$$\text{Entropy}\ (S_{Weak}) = 0.811$$

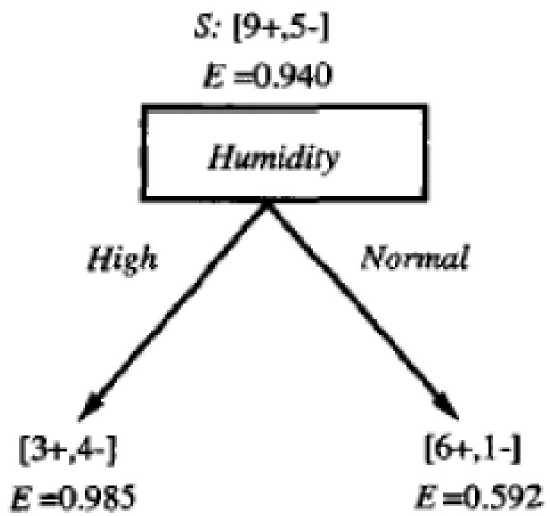
$$\text{Entropy}\ (\ S_{Stong}\ ) = -\,3/6\ \log_2 3/6 \quad -\,3/6\ \log_2 3/6$$

$$\text{Entropy}\ (\ S_{Stong}\ ) = 1$$

Decision Tree

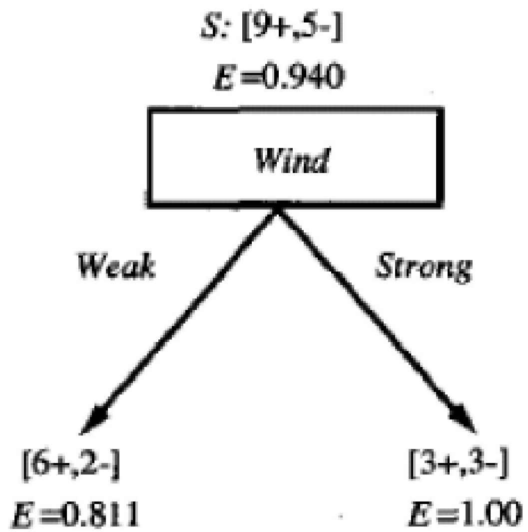
Which Attribute is the Best Classifier? Information Gain

We calculate the Info Gain for each attribute and select the attribute having the highest Info Gain



Gain (S, Humidity)

$= .940 - (7/14).985 - (7/14).592$
$= .151$



Gain (S, Wind)

$= .940 - (8/14).811 - (6/14)1.0$
$= .048$

Decision Tree

Which Attribute is the Best Classifier? Information Gain

The information gain obtained by separating the examples according to the attribute Temperature is calculated as:

$$Gain\,(S, Temp) = Entropy\,(S) - 4/14\; Entropy\,(S_{Hot}) - 6/14\; Entropy\,(S_{Mild}) - 4/14\; Entropy\,(S_{Cool})$$

$$\text{Gain}\,(\text{S}, \text{Temp}) = 0.029$$

Decision Tree

Alternate Formula

The formula can be converted from $\log_2$ to $\log_{10}$

$$\begin{aligned} \log_x(M) &= \log_{10}M \,.\, \log_x 10 \\ &= \log_{10}M/\log_{10}x \end{aligned}$$

Hence $\log_2(Y) = \log_{10}(Y)/\log_{10}(2)$

Decision Tree

Example

Which attribute should be selected as the first test?

$$Gain(S, Outlook) = 0.246$$

$$Gain(S, Humidity) = 0.151$$

$$Gain(S, Wind) = 0.048$$

$$Gain(S, Temperature) = 0.029$$

"Outlook" provides the most information

Decision Tree

Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
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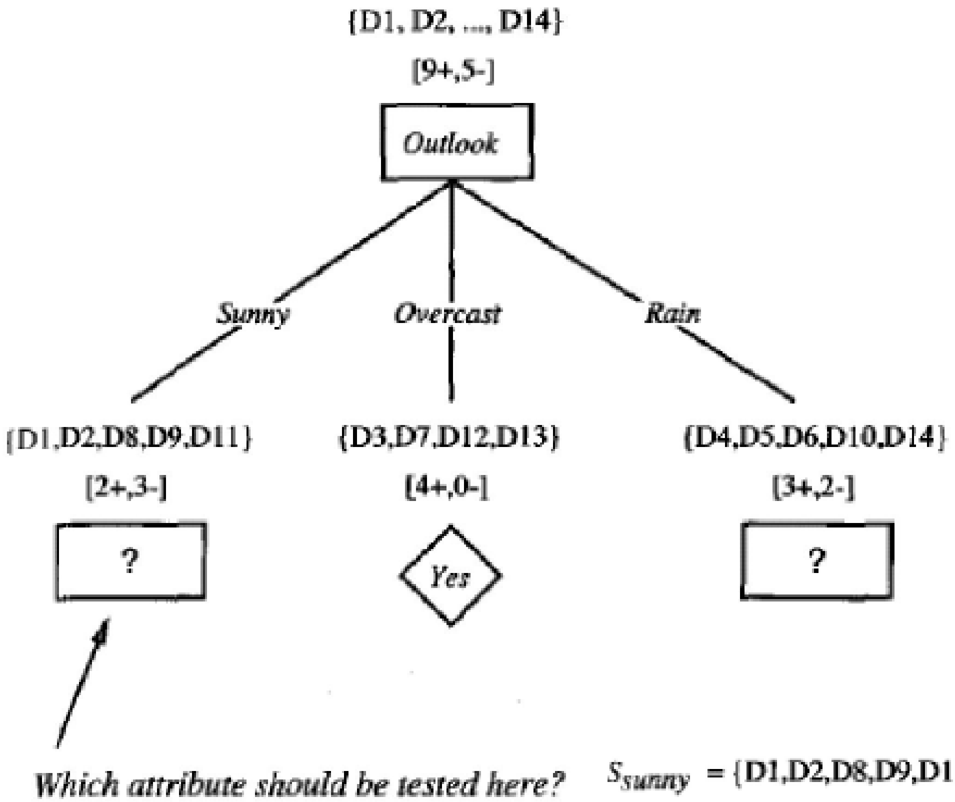
Decision Tree



S_{sunny} = {D1,D2,D8,D9,D11}

$Gain(S_{sunny}, Humidity) = .970 - (3/5)\,0.0 - (2/5)\,0.0 = .970$

$Gain(S_{sunny}, Temperature) = .970 - (2/5)\,0.0 - (2/5)\,1.0 - (1/5)\,0.0 = .570$

$Gain(S_{sunny}, Wind) = .970 - (2/5)\,1.0 - (3/5)\,.918 = .019$

Decision Tree

The information gain obtained by separating the examples according to the attribute Temperature is calculated as:

$$\begin{aligned} Gain\,(S, Temp) \;=\; & Entropy\,(S) - 4/14\; Entropy\,(S_{Hot}) \\ & - 6/14\; Entropy\,(S_{Mild}) \\ & - 4/14\; Entropy\,(S_{Cool}) \end{aligned}$$

$$\mathrm{Gain}\,(\mathrm{S}, \mathrm{Temp}) \;=\; 0.029$$

Decision Tree

Example

The process of selecting a new attribute is now repeated for each (non-terminal) descendant node, this time using only training examples associated with that node

Attributes that have been incorporated higher in the tree are excluded, so that any given attribute can appear at most once along any path through the tree

Decision Tree

Example

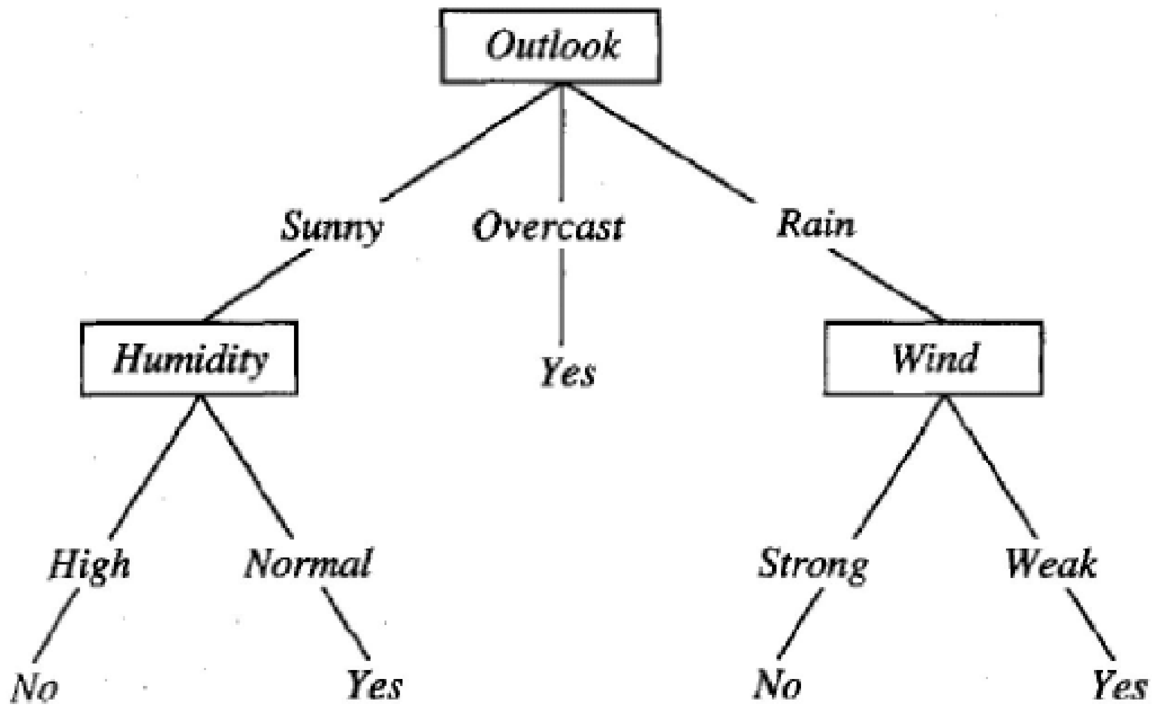
This process continues for each new leaf node until either:

1. **Every attribute has already been included along this path through the tree**

2. The training examples associated with a leaf node have zero entropy

Decision Tree

Example



Decision Tree

From Decision Trees to Rules

Next Step: Make rules from the decision tree

After making the identification tree, we trace each path from the root node to leaf node, recording the test outcomes as *antecedents* and the leaf node classification as the *consequent*

For our example we have:

If the Outlook is Sunny and the Humidity is High then No
If the Outlook is Sunny and the Humidity is Normal then Yes
...